Lecture 20: Polar Coordinates

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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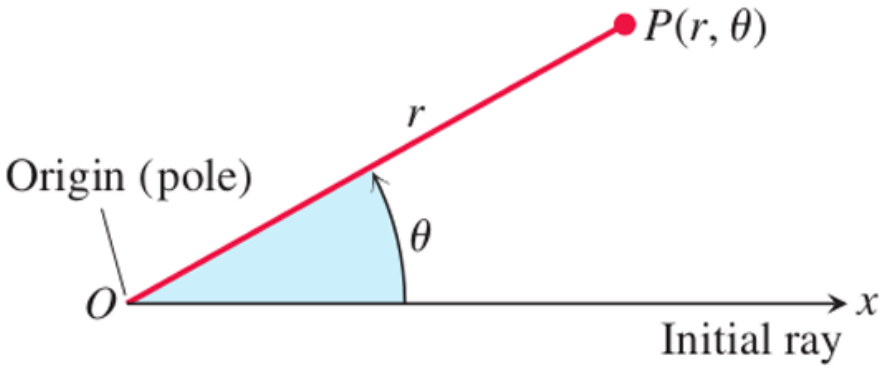
Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand the definition of polar coordinates.
- ☐ Identify the geometric meaning of polar equations and recognize common polar graphs.
- ☐ Sketch graphs of polar equations.
- ☐ Convert equations between polar and Cartesian coordinate systems.

Polar Coordinates

Many problems in mathematics are easier to solve if we choose a suitable coordinate system. Previously, we use Cartesian coordinate. In this section, we study polar coordinates.



Fix an origin O (pole) and an initial ray from O. Then each point P can be located by assigning it to a polar coordinate pair (r, θ) where

$$r = \text{ the directed distance from } O \text{ to } P,$$
$$\theta = \text{ the directed angle from the initial ray to ray } OP.$$

Polar Coordinates

Example: Find all the polar coordinates of the point $P(2, \pi/6)$.

Polar Equations and Graphs

In the polar coordinate system, the equations

(i) $r = a$ is a circle with radius $|a|$ centered at O, and

(ii) $\theta = \theta_0$ is a line through O that makes an angle θ_0 with the initial ray.

Example:

(i) $r = 1$ and $r = -1$ are equations for the circle of radius 1 centered at O.

(ii) $\theta = \pi/6$, $\theta = 7\pi/6$ and $\theta = -5\pi/6$ are equations for a line.

Polar Equations and Graph

Example: Graph the sets of points whose polar coordinates satisfy the following conditions.

(a) $1 \leq r \leq 2$ and $0 \leq \theta \leq \frac{\pi}{2}$

(b) $-3 \leq r \leq 2$ and $\theta = \frac{\pi}{4}$

(c) $\frac{2\pi}{3} \leq \theta \leq \frac{5\pi}{6}$ (no restriction on r)

Relating Polar and Cartesian Coordinates

In both systems, if we place the two origins together, then the initial polar ray be the positive x-axis and the ray $\theta = \frac{\pi}{2}$, $r > 0$ becomes the positive y-axis. The two coordinate systems are then related by

- $x = r\cos\theta$;
- $y = r\sin\theta$;
- $r^2 = x^2 + y^2$;
- $\tan\theta = \frac{y}{x}$.

Example:

$r\cos\theta = 2$

$r^2\cos\theta\sin\theta = 4$

$r^2\cos^2\theta - r^2\sin^2\theta = 1$

$r = 1 + 2r\cos\theta$

$r = 1 - \cos\theta$

Relating Polar and Cartesian Coordinates

Find a polar equation for the circle $x^2 + (y - 3)^2 = 9$.

Relating Polar and Cartesian Coordinates

Replace the following polar equations by equivalent Cartesian equations and identify their graphs.

(a) $r\cos\theta = -4$

(b) $r^2 = 4r\cos\theta$

(c) $r = \frac{4}{2\cos\theta - \sin\theta}$

(d) $r = 2\cos\theta$